

Functions Extension · 64 Cages — English Worksheet

Topic: One function, three loops · **Course:** Functions & Games · **Level:** Extension (after helper functions) · **Time:** about 45 minutes

Last time one function built one cage. This time the same function builds **64**, stacked in a **4 × 4 × 4** tower. Every cage holds an animal and stands on grass.

Your cage from last lesson:

```
def make_cage(x: number, y: number, z: number, animal: number):
    blocks.fill(OAK_FENCE,
        pos(x, y, z),
        pos(x + 4, y + 3, z + 4), FillOperation.HOLLOW)
    blocks.fill(GRASS,
        pos(x, y, z),
        pos(x + 4, y, z + 4), FillOperation.REPLACE)
    mobs.spawn(animal, pos(x + 2, y + 1, z + 2))
```

One cage is **5 wide, 4 tall, 5 deep**.

1 · Predict 🧠

Read each one. Write the number **and the reason**.

```
for col in range(4):
    make_cage(col * 6, 0, 0, LLAMA)
```

How many cages? Which way do they run?


```
for row in range(4):
    for col in range(4):
        make_cage(col * 6, 0, row * 6, LLAMA)
```

How many cages now? What shape do they make?


```
for level in range(4):  
    for row in range(4):  
        for col in range(4):  
            make_cage(col * 6, level * 5, row * 6, LLAMA)
```

How many cages? Which loop lifts them off the ground?

A cage is 4 tall. The levels are 5 apart. What would happen if you changed `level * 5` to `level * 3`?

2 · Spot the Bug 🐛

Each one runs. Each one is wrong. Fix it, then say why.

Bug A — This should build 64 cages. You count 16.

```
for row in range(4):  
    for col in range(4):  
        make_cage(col * 6, 0, row * 6, LLAMA)
```

Write the fix:

Why was it wrong?

Bug B — Every animal disappears the moment the cage appears.

```
def make_cage(x: number, y: number, z: number, animal: number):  
    mobs.spawn(animal, pos(x + 2, y + 1, z + 2))  
    blocks.fill(OAK_FENCE,  
        pos(x, y, z),  
        pos(x + 4, y + 3, z + 4), FillOperation.HOLLOW)
```

Write the fix:

Why was it wrong?

Bug C — All 64 cages land in one flat layer, cutting through each other.

```
for level in range(4):
    for row in range(4):
        for col in range(4):
            make_cage(col * 6, 0, row * 6, LLAMA)
```

Write the fix:

Why was it wrong?

3 · Read the Grid

Each loop owns one direction.

loop	it changes	direction
col	x	across
level	y	up
row	z	away from you

Fill in the missing numbers for `make_cage(col * 6, level * 5, row * 6, LLAMA)` :

col	level	row	x	y	z
0	0	0	0	0	0
3	0	0			
0	2	0			
1	1	1			
3	3	3			

The last row is the far top corner. How far is that cage from where you stand?

4 · Build It

Stand somewhere with open space above you and behind you. The tower needs about **20 blocks** in each direction.

Fill the blanks and run it:

```
def make_cage(x: number, y: number, z: number, animal: number):
    blocks.fill(OAK_FENCE,
        pos(x, y, z),
        pos(x + 4, y + 3, z + 4), FillOperation.HOLLOW)
    blocks.fill(GRASS,
        pos(_____, _____, _____),
        pos(x + 4, _____, z + 4), FillOperation.REPLACE)
    mobs.spawn(_____, pos(x + 2, y + 1, z + 2))

for level in range(____):
    for row in range(____):
        for col in range(____):
            make_cage(col * 6, level * _____, row * 6, _____)
```

Your tower must have:

#	Check
1	64 cages, 4 across, 4 up, 4 deep
2	an animal alive inside every cage
3	grass on the floor of every cage
4	no cage cutting through another cage

Once the 64 stand up, keep going. Different animals on each level, or a gate you can walk through.

5 · Show Your Work 📷 🎥

Run your code and walk around the finished tower.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1. Your code on screen, and explain what each of the three loops does.
2. The tower in the world, and count a row of cages out loud.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit

Send your video on KakaoTalk.